

Introduction to Macroeconomics (S102016CR)

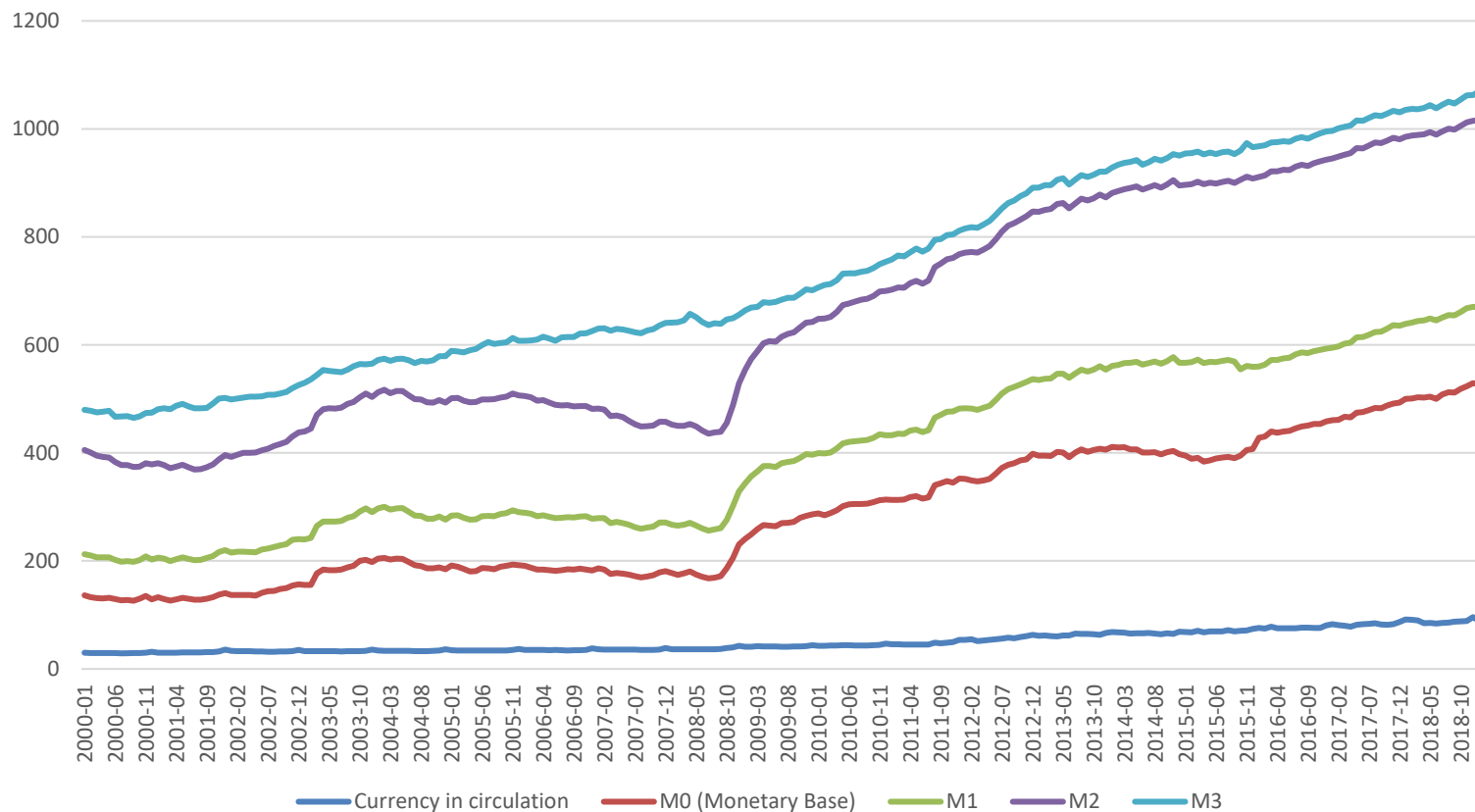
Spring Semester

<https://moodle.unige.ch/course/view.php?id=6618>

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Friday session:
For Weeks 5 and 6

What happened with money aggregates during the 2007-2009 financial crisis?



- Why M0-M2 go swiftly upwards in 2008-2009, and M3 stays on trend (or actually goes slightly downwards)?
- Why does the upwards trend continue (hint 2011-2014)?

The money multiplier more formally:

- Commercial banks can only lend a fraction $x=(1-r)$ of the deposits. Thus, if this takes place n times, the total money supply will be given by:

$$MS = M0*(1+x +x^2+x^3+... +x^n)$$

- This is a geometric series that converges to a finite number: $1/(1-x)$ when $x < 1$ and n tends to infinity.
- To see this, rewrite the total amount:

$$MS = M0 \lim_{n \rightarrow \infty} (1 + x + x^2 + x^3 + \dots + x^n)$$

- Multiply each side of the equality by $(1-x)$ to obtain:

$$(1 - x)MS = M0 \lim_{n \rightarrow \infty} (1 - x^{n+1}) = M0$$

- Rearrange to obtain:

$$MS = M0 \left(\frac{1}{1-x} \right) = M0 \left(\frac{1}{r} \right)$$

Local currencies

- Complements not substitutes to official currency
- Used voluntarily at a fixed exchange rate and may be convertible or not
- More than 5000 examples around the world (WIR was introduced in the 30s in Switzerland, the leman in 2015 and the Farinet had a short existence in Valais between 2017-2020)
- But even mileage point cards can be thought of as local currency or time-banks for babysitting services, for example
- They are supposed to stimulate local economic activity
- They should be considered money as they are a medium of exchange and provide a unit of account. Not all are adequate to store value (savings)
- Their value is limited geographically or to a few goods or types of markets
- They are more likely to be counterfeit
- Relative more problems with tax evasion (but this is a general problem with currency and is the reason why some lobby for the elimination of currency)
- Rise of the use of digital currency makes physical currencies less attractive

Are cryptocurrencies money?

- Cryptocurrencies can be considered digital cash
- A digital record gives their existence
- But contrary to bank accounts, the digital record is not in a financial institution but stored in a blockchain in many computers. It is a totally decentralized system
- Its value, like with fiat money, depends on the faith that individuals have in the value that the cryptocurrency can procure in the future (and, therefore, how many people will accept it as a means of payment).
- In some countries with weak and unreliable central banks, they can provide a reliable alternative to the central bank.
- But in most countries, they are not commonly accepted as a medium of exchange (one condition for the definition of money)
- Also, the volatility of some of these cryptocurrencies has made them unreliable as a store of value (one condition for the definition of money)
- So, ok as a unit of account, but not as a medium of exchange or store of value.

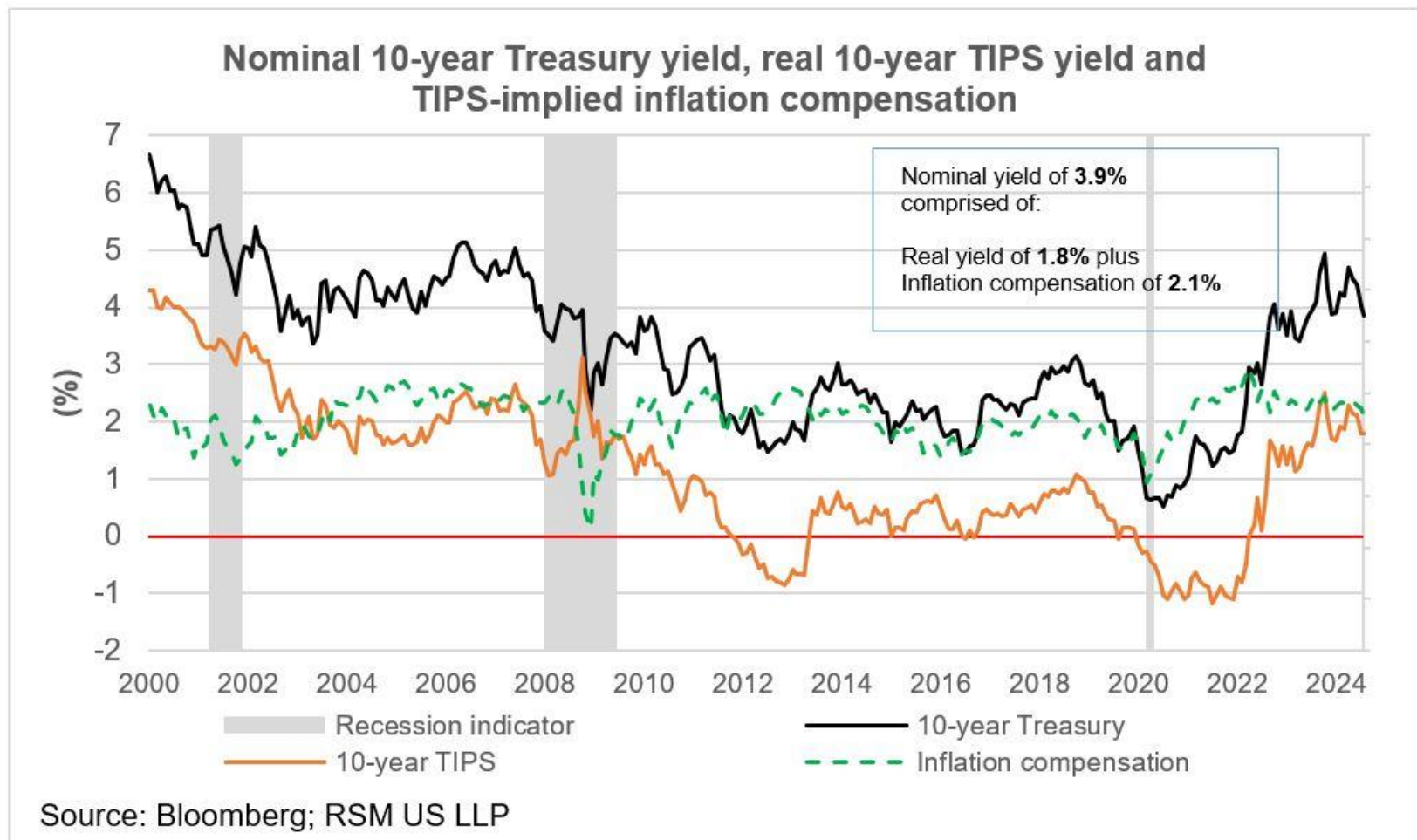
What do shortages of toilet paper during COVID and bank runs have in common?

Bank Runs

	Others withdraw	Others keep money in the bank
You withdraw your money	Bank run: We race to see who gets their money out first ✓✓	You don't earn interest Others definitely lose their savings
You keep your money in the bank	You definitely lose your savings Others don't earn interest	Stability: We both earn interest ✓✓

How to calculate expected inflation?

- We could do surveys, but people are likely to answer randomly without putting too much thought. Alternatively, we can look at what markets have to say:



Fisher parity and money neutrality

Suppose there is an increase in expected inflation. What happens with the real and nominal interest rates ex-ante?

Suppose there is an unexpected increase in inflation. What happens with the real and nominal interest rates ex-post?

The Exam

Material covered

- All that has been seen in class or seminar
- The companion chapters in M&T or K&W if the material was covered in class or seminar
- No surprises in terms of material covered
- No differences between the English and French exams (apart from language).

Type of exam

- 2 hours
- 30 MCQ with 4 possible answers. No penalties → Answer all questions
- There will be 6 to 8 questions on the first part of the course, 6 to 8 questions on the second part, 6 to 8 questions on the third part, and 6 to 8 questions on the fourth part.
- Over the entire exam
 - 1/3 of the questions will require some simple calculus
 - 2/3 of the questions will be analytical
- A two-sided A4 page of personal notes and a non-programmable calculator are allowed

Examples of exam questions for weeks 5 and 6

votamatic.unige.ch

code d'accès: JTML



<https://votamatic.unige.ch/form/2ly4vlyYM7vsK6qSuRMD>